PHYSICS LAB 3 (21ST OF MAY 2025):

LORENTZ FORCE

Classical electrodynamics of charged particles in a magnetic field

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1 Introduction

Inspired by J. J. Thomson's original cathode-ray studies, in this lab we studied electrons moving through a homogeneous magnetic field inside a cathode-ray tube. We first mapped the field generated by a single coil and by a Helmholtz pair, then varied field strength, electron speed and beam orientation to observe the resulting cyclotron orbits. By analysing these trajectories with the Lorentz-force law and comparing the data with predictions from the Biot–Savart law and classical dynamics, we determined the electron's specific charge e/m and compared our experimental result with the known specific charge of the electron.

2 Generation and measurement of a magnetic field

2.1 Magnetic field generated by a single coil

1.

Using the Biot-Savart Law we computed the value of the magnetic field along the axis passing through the center of a coil created by that coil. The formula we got is:

$$B_z(z) = \frac{\mu_0 N I R^2}{2(R^2 + z^2)^{3/2}}$$

where:

- μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ T · m/A),
- N is the number of turns in the coil,
- I is the current in the coil (A),
- R is the radius of the coil (m),
- z is the distance along the axis from the center of the coil (m).

For our measurement, given that the Teslameter had a total length of 28 cm, we considered that a $\Delta z = 2$ cm would be sufficient for a smooth enough curve (giving 14 points).

In order to pinpoint the position of the detector on the Teslameter, we used the fact that the maximum magnetic field generated by the coil is found in the center of the coil, hence when we noticed a peak in the magnetic field we knew that the detector was located in the center of the coil. Next, we set that point to 12 on the scale of the Teslameter (right in the middle of the axis so that we can take as many values on the left on the coil as on the right)

We fixed the intensity to 1.01 Amperes

This is a table of the data we collected using the Teslameter:

Table 1: B as a function of z

z (cm)	B (mT)
-12	0.25
-10	0.3
-8	0.37
-6	0.43
-4	0.49
-2	0.53
0	0.55
2	0.53
4	0.5
6	0.45
8	0.38
10	0.32
12	0.26
14	0.22

Using the following code in Matlab we have plotted the theoretical curve for the magnetic field along the axis passing through the center of a coil created by that coil:

```
% Parameters
mu0 = 4 * pi * 1e-7;  % Permeability of free space (T·m/A)
R = 0.1475;           % Radius of the coil in meters
N = 124;              % Number of turns
I = 1.01;             % Current in Amperes

% z-axis range (e.g., -0.1 m to 0.1 m)
z = linspace(-0.14, 0.14, 500);  % meters

% Magnetic field along the axis
Bz = (mu0 * N * I * R^2) ./ (2 * (R^2 + z.^2).^(3/2)) * 1e3 %convert to mT;

% Plot
figure;
plot(z, Bz, 'b-', 'LineWidth', 2);  % theoretical curve
hold on;
```

Figure 1: Matlab code for B(z)

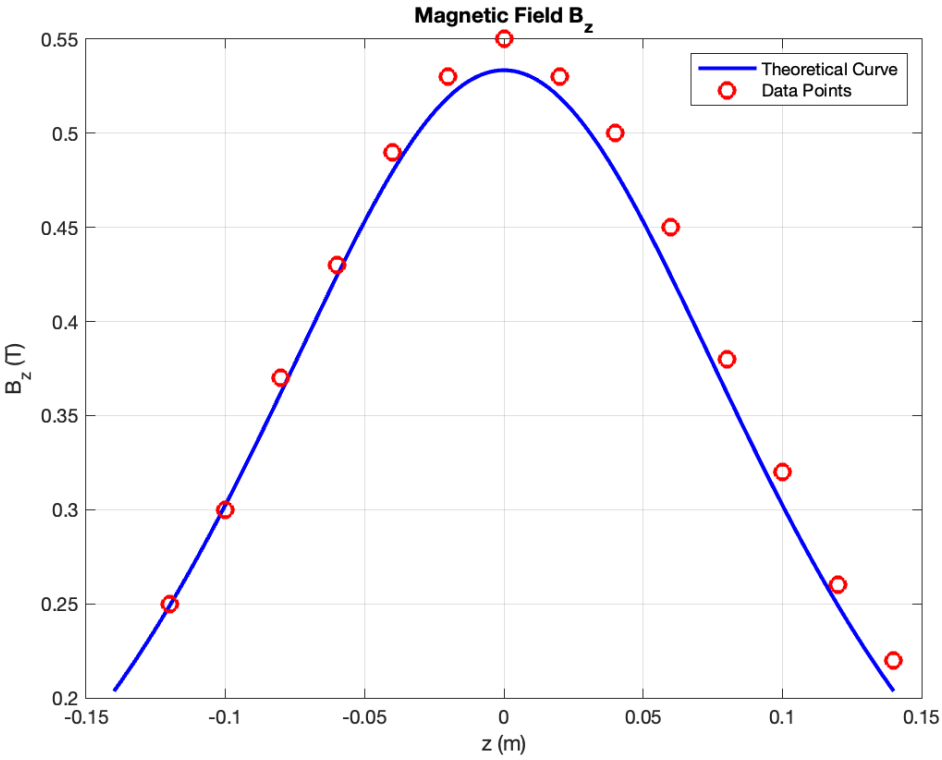

Figure 2: Theoretical vs Experimental B(z)

Furthermore, we added the set of discrete points we measured during the lab, thus being able to compare our experimental results to the theoretical prediction.

2. The length of w is about 1.5 cm or 15 mm so the ratio w/R is approximately 0.1 << 1, so theoretically the width of the coil changes how the theoretical curve should look like but since the ratio is very small compared to 1 it can be neglected.

2.2 Magnetic field generated by a pair of coils: Helmholtz

To easily determine which configuration we are in, we left the Teslameter in the area betwen the coils. We imagined theoretically that in the Helmholtz configuration, the intensities having the same direction, would create magnetic field which being in the same direction would have additive superpositon. Conversely, in the case of the Anti-Helmholtz configuration, the magnetic fields would oppose each other, the superposition being destructive and the value of the magnetic field nearing 0 in the center of the system.

In order to achieve Helmholtz configuration we need the two currents to have the same direction, so we first arbitrarily decided to connect the + voltage to 1 and so the current would flow from 1 to

2 as a convention. Looking at the coil this implies a counterclockwise flow given our convention. For Helmholtz, we would need the same direction of the current which means connecting + to 4 and 3 to -. Therefore, the sequence we used for the Helmholtz configuration is:

$$+ \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow -$$

Using the same technique of taking measurements we got the following values.

Table 2: B as a function of z

z (cm)	B (mT)
-14	0.52
-12	0.6
-10	0.68
-8	0.73
-6	0.77
-4	0.78
-2	0.78
0	0.78
2	0.78
4	0.78
6	0.76
8	0.73
10	0.68
12	0.62

Using the superposition principle of the magnetic field we get:

$$B(z) = B_{coil1}(z) + B_{coil2}(z)$$

$$B(z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2 + \left(z - \frac{z_0}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z + \frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

and in our case $z_0 = R$. Now, we plotted the graph of this function in the range from which we took experimental data points and compared the results.

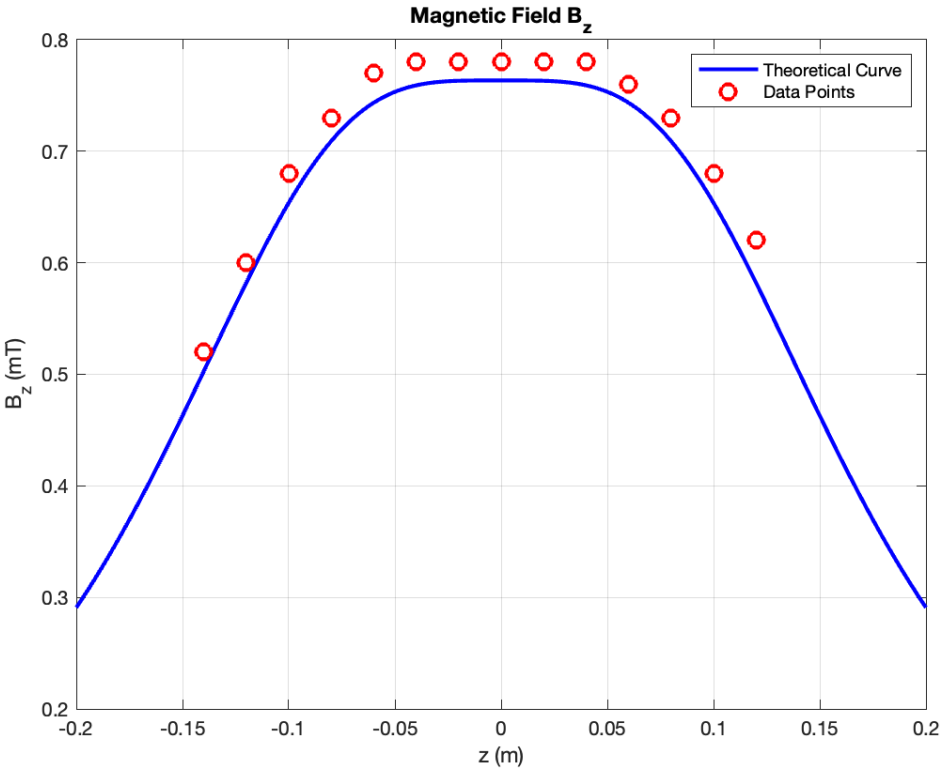

Figure 3: Plot of the magnetic field in the Helmholtz set-up

Before, we computed the expression for the magnetic field in this scenario:

$$B(z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2 + \left(z - \frac{z_0}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z + \frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

Now, we compute $B(-z)$:

$$B(-z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2 + \left(-z - \frac{z_0}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(-z + \frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

Using the identity $(-a)^2 = a^2$, we simplify:

$$B(-z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2 + \left(z + \frac{z_0}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z - \frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

$$\boxed{B(-z) = B(z)}$$

Hence, we can conclude that this is an even function in the variable z. Furthermore, this is confirmed by the graph where the z = 0 axis represents a mirror axis for the graph of B(z).

We notice that our measurement is very close to the theoretical prediction though slightly higher in value.

In the next graph we plotted the curves of B(z) for varying values of z_0 as mentioned in the lab guidelines.

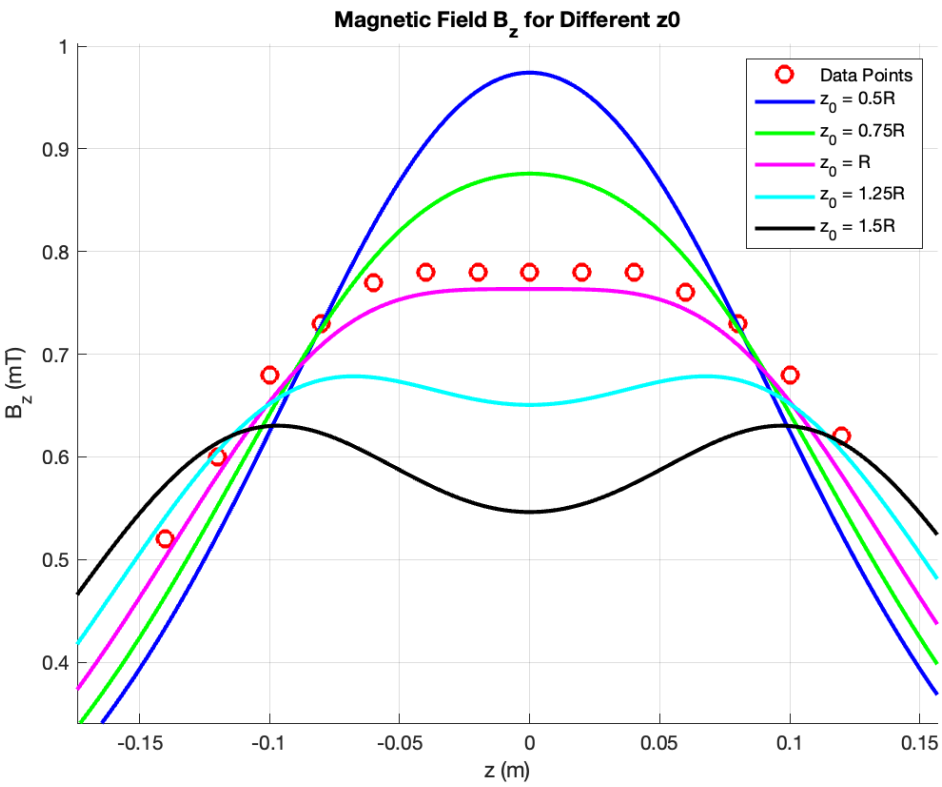

Figure 4: Magnetic field curve with varying z0

We notice that in the case where z0 = R, we experience the largest homogeneous zone, as it happens that the two magnetic fields are almost complementary around the z=0 region, more specifically for $z \in [-4, 4]$. This makes it an ideal candidate for our next experiment where we will benefit from the quasi-constant magnetic field.

2.3 Magnetic field generated by a pair of coils: Anti-Helmholtz

As mentioned before, in this configuration we have to do the opposite, create an opposing current with the second coil, hence generating an opposing magnetic field. To achieve the inversely flowing current, we will switch the two terminals (3 and 4) such that current flows clockwise (from our perspective and according to the convention made).

Explicitely, this means that our circuit looks like this:

$$+ \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow -$$

The results measured in this configurations are presented in the following table:

Table 3: B as a function of z

z (cm)	B (mT)
-14	0.33
-12	0.37
-10	0.4
-8	0.38
-6	0.33
-4	0.24
-2	0.13
0	0
2	0.12
4	0.22
6	0.31
8	0.35
10	0.37
12	0.34

Using the expression deduced in the previous configuration, we just have to change I to -I as the direction of the current has been changed. Thus, we get the following formula for B(z):

$$B(z) = \frac{\mu_0 N I R^2}{2} \left[-\frac{1}{\left(R^2 + \left(z - \frac{z_0}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z + \frac{z_0}{2}\right)^2\right)^{3/2}} \right]$$

Plotting is along our data we see that they are very similar:

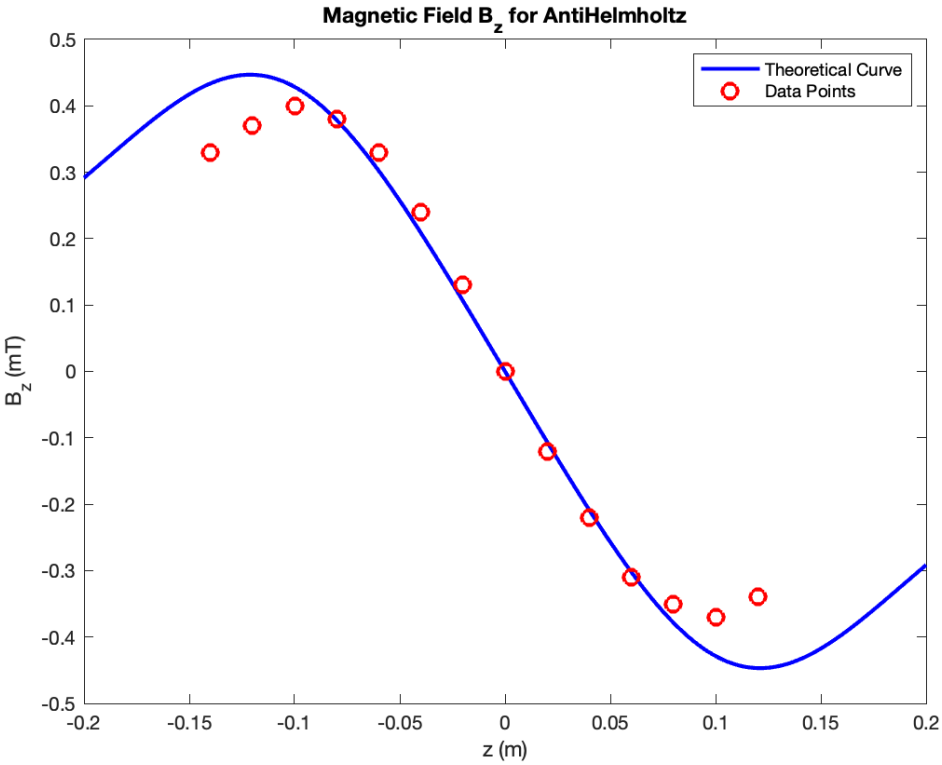

Figure 5: Antihelmholtz configuration B(z)

From the graph as well as from the expression of B(z) we can conclude that it is an odd function.

$$B(-z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2+\left(-z-\frac{z_0}{2}\right)^2\right)^{3/2}} - \frac{1}{\left(R^2+\left(-z+\frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

Using the identity $(-a)^2 = a^2$, we simplify:

$$B(-z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2+\left(z+\frac{z_0}{2}\right)^2\right)^{3/2}} - \frac{1}{\left(R^2+\left(z-\frac{z_0}{2}\right)^2\right)^{3/2}}\right]$$

$$B(-z) = -\left(\frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2+\left(z-\frac{z_0}{2}\right)^2\right)^{3/2}} - \frac{1}{\left(R^2+\left(z+\frac{z_0}{2}\right)^2\right)^{3/2}}\right]\right)$$

$$\boxed{B(-z) = -B(z)}$$

Additionally, the value of z_* for which $B(z_*) = 0$ is $z_* = 0$.

A source of error we found worth mentioning was the movement of the Teslameter rod, not always remaining perfectly perpendicular to the plane of the coil, which affects the values measured.

3 Specific charge of the electron

3.1 Electron beam tube setup

Following our experiments on the coil configurations in the previous part, we concluded that the Helmholtz configuration with $z_0 = R$ gave us the largest homogeneous zone for the magnetic field, therefore we will use this configuration to set up the electron beam tube.

The electron beam tube was installed between the pair of coils, with an ammeter connected in series and a voltmeter connected in parallel with the acceleration voltage to measure the pairs $(V_{acc}, I$ for the next part.

3.2 Qualitative observations

We first notice that the the electron beam follows a helix pattern, aligned with the z-axis[1], as shown in the figure below.

[1] We use the coordinate system defined in Figure 5 in Appendix 1.

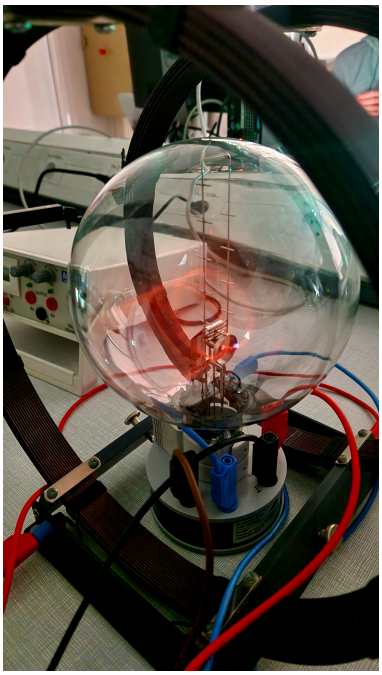

Figure 6: Cyclotron motion observed in Electron beam tube

When the electron gun is aligned with the x-axis, the motion of the electrons is circular, in a plane parallel to the coils.

Cyclotron orbit when varying V_{acc}: We observe experimentally that the radius of the orbit increases with the acceleration voltage.

Cyclotron orbit when varying the magnetic field: We observe experimentally that the radius of the orbit decreases as the magnitude of the magnetic field increases.

According to Appendix 2, in a homogeneous magnetic field B, the electrons follow a cyclotron orbit of radius:

$$R_c = \frac{m}{e}\frac{v_\perp}{B}$$

This equation confirms our observations on the relation between the magnetic field and the radius of the orbit. Furthermore, by the kinetic energy theorem, we know that the net work done on a particle equals the change in its kinetic energy. The work done by the electric field on an electron of charge e moving through a potential difference V_{acc} is $W = eV_{acc}$. The electron gains kinetic energy $\frac{1}{2}mv_\perp^2$. Thus:

$$eV_{acc} = \frac{1}{2}mv_\perp^2$$

$$v_\perp = \sqrt{\frac{2eV_{acc}}{m}}$$

By combining those results we get:

$$R_c = \frac{m}{eB}\sqrt{\frac{2eV_{acc}}{m}}$$

This confirms our observations on the relation between V_{acc} and the radius of the orbit. We can rewrite this result in another form:

$$2V_{acc} = \frac{e}{m}(R_c B)^2$$

3.3 Quantitative measurements and determination of $\frac{e}{m}$

We can now, by measuring several pairs of (V_{acc}, I) values that maintain a given R_c constant, then computing the magnetic field using I and plotting $2V_{acc}$ as a function of $(R_c B)^2$, obtain the slope given by $\frac{e}{m}$. Recall that the magnetic field corresponding to I in our Helmhotz configuration where $z_0 = R$ is given by:

$$B(z=0) = \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 N I}{R}$$

with $N = 124$, $R = 147.5mm$ and $\mu_0 = 4\pi 10^{-7} H.m^{-1}$.

We will take our measurements for different fixed values of R.

$R_c(cm)$	$V_{acc}(V)$	$I(A)$	$B(T)$	$(R_c B)^2 \times 10^9 (m^2.T^2)$
2	280	2.62	0.00198	1.569
2	291	2.57	0.00194	1.600
2	253	2.09	0.00158	0.998
2	217	1.88	0.00142	0.808
3	267	0.77	0.00058	0.305
3	316	1.11	0.00084	0.634
3	447	2.72	0.00206	3.805
4	335	1.63	0.00123	2.429
4	360	1.73	0.00131	2.736
4	400	1.86	0.00141	3.163
4	422	1.94	0.00147	3.441
4	470	2.09	0.00158	3.994
4	533	2.22	0.00168	4.506

Table 4: Data for fixed R_c values

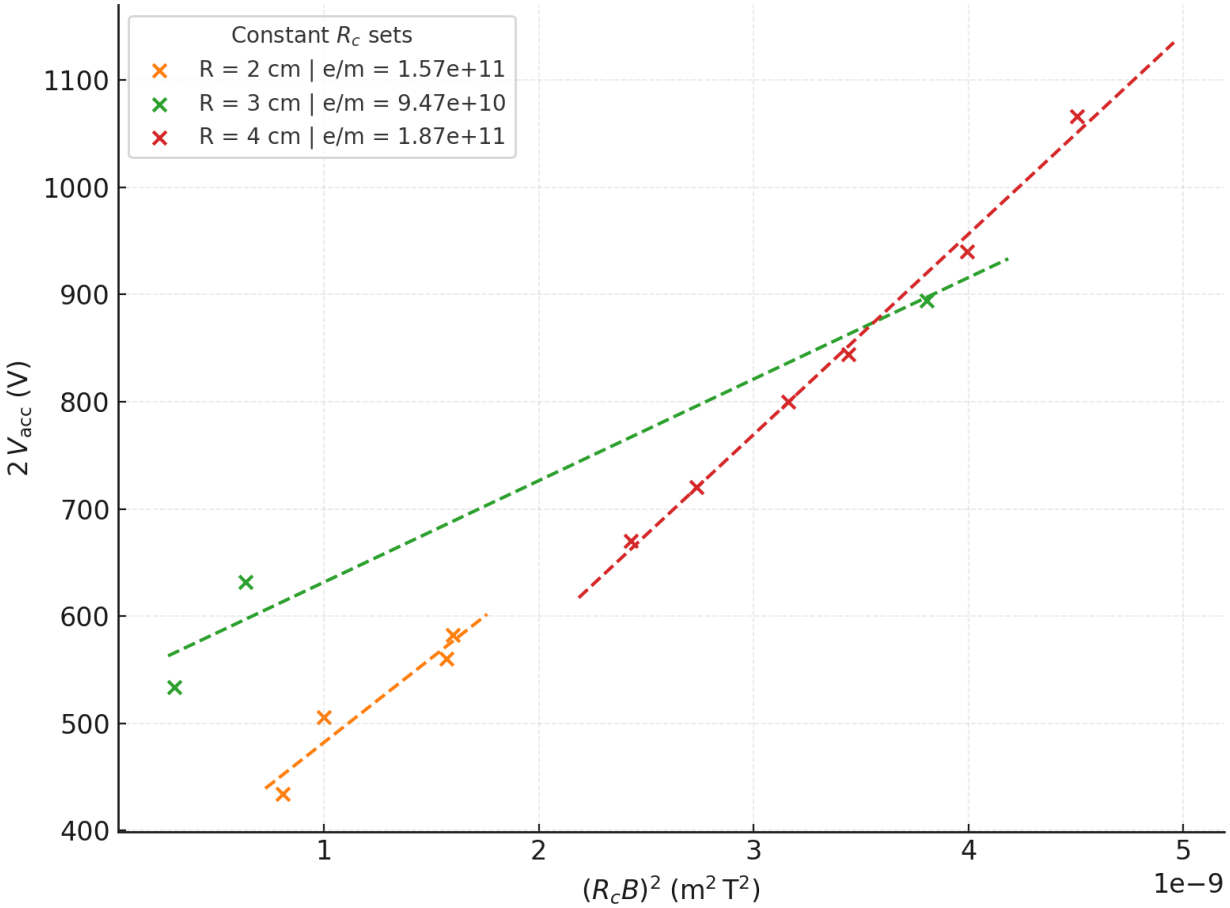

Figure 7: Plot of $2V_{acc}$ as a function of $(R_cB)^2$

As seen on the plot, the values of $\frac{e}{m}$ we obtained for each cyclotron orbit radius R_c are (in C/kg):

- $R_c = 2cm : 1.57 \times 10^{11}$
- $R_c = 3cm : 9.47 \times 10^{10}$
- $R_c = 4cm : 1.87 \times 10^{11}$

We see that the slopes derived from each cyclotron orbit radius are all of the 10^{11}C/kg order. We compute the average value and the standard deviation of e/m:

$$\frac{\bar{e}}{m} = \frac{1.57 \times 10^{11} + 9.47 \times 10^{10} + 1.87 \times 10^{11}}{3} = 1.46 \times 10^{11} C/kg$$

$$\sigma(\frac{e}{m}) = \sqrt{\frac{\Sigma(\frac{e}{m\,i} - \frac{\bar{e}}{m})^2}{3}} = 3.84 \times 10^{10} C/kg$$

The standard deviation gives us a relative uncertainty of $\approx$ 26% of the average. Knowing the accepted values of $e = 1.60217662 \times 10^{-19} C$ and $m = 9.10938356 \times 10^{-31} kg$, we can calculate the

accepted value of the specific charge of the electron:

$$\frac{e}{m}^{\star} \approx 1.76 \times 10^{11}$$

We compute the relative error to compare our experimental value and the accepted value:

$$\frac{1.76 \times 10^{11} - 1.46 \times 10^{11}}{1.76 \times 10^{11}} \approx 17\%$$

We can firstly observe that our values for the largest radius we measured ($R_c = 4cm$) are the closest to the accepted value of the specific charge. This can underline an error of measurement and uncertainties with the smaller values of R_c, as for these smaller values it was harder to accurately measure the orbit. We also found uncertainties about the acceleration voltage which wasn't fully stable during the experiment, as well as the the magnetic field.

We can now check that the units are consistent to ensure the validity of our experiment, using the S.I units: We know e/m is in C/kg, and $e/m = \frac{2V_{acc}}{(R_c B)^2}$. V_{acc} is in Volts, and $[V] = \frac{kg.m^2}{A.s^3}$. R_c is in meters and B is in Tesla, $[T] = \frac{A.s}{kg}$. So $(R_c B)^2$ is in $\frac{m^2.kg^2}{A^2.s^4}$ and $\frac{2V_{acc}}{(R_c B)^2}$ is in $\frac{kg.m^2}{A.s^3}.\frac{A^2.s^4}{m^2.kg^2} = \frac{A.s}{kg} = C/kg$. The units are consistent.

To conclude, we found an experimental value of the specific charge of the electron of $1.46 \times 10^{11} \pm 3.84 \times 10^{10} C/kg$, with a relative error of 17% with the accepted value $\frac{e}{m}^{\star} = 1.76 \times 10^{11}$. Our experimental interval does contain the accepted value of the specific charge of the electron, so we can say that our experiment was successful, however we must attribute our large relative error to measuring errors:

- We measured the magnetic field by measuring the current passing through the coils, our values for the current were not stable and the ammeters can have a small uncertainty of measurement. Moreover, the placement of the electron beam tube wasn't very precise and a small misalignment with z=0 could have impacted our results.
- Since $e/m \propto 1/R_c^2$, small measurement errors from our part with the built-in ruler of the electron beam tube can have a bigger impact in our results.
- There is an uncertainty relative to the acceleration voltage, from the power supply.

4 Charged particles in an inhomogeneous magnetic field : the case of a magnet

For this configuration, we turned off the current passing through the coils to eliminate the magnetic field. We then brought a magnet close to the electron beam and instantly observed that the beam was repelled by the magnet, as seen in the following figure.

We observed that adjusting the Wehnelt voltage altered the shape of the beam, as a lower voltage gave a larger beam and a higher voltage gave a thinner and more visible beam.
The magnet deflects the beam because the electrons, while moving, feel a sideways Lorentz force from the magnetic field. This changes their direction, giving the visible deflection observed.
The case of the electron beam being deflected by a magnet is similar to the Northern Lights phenomenon. Indeed, the Earth's magnetic field guides charged particles in its atmosphere along its field lines, causing this phenomenon.

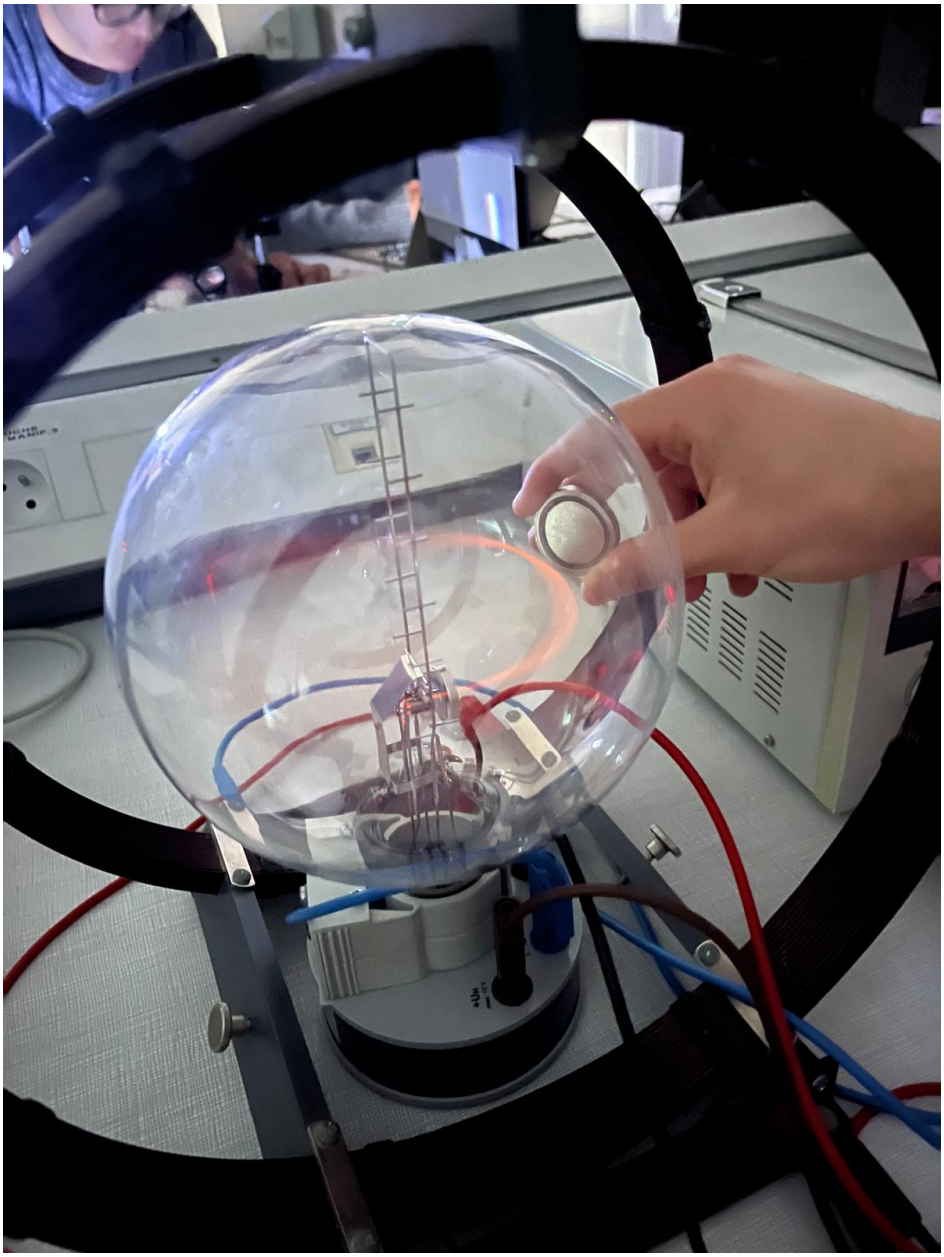

Figure 8: Electron beam when influenced by a magnet